



CYBERITH **IK**

Basics & General Information

Inverse Kinematics system based on Final IK VRIK by



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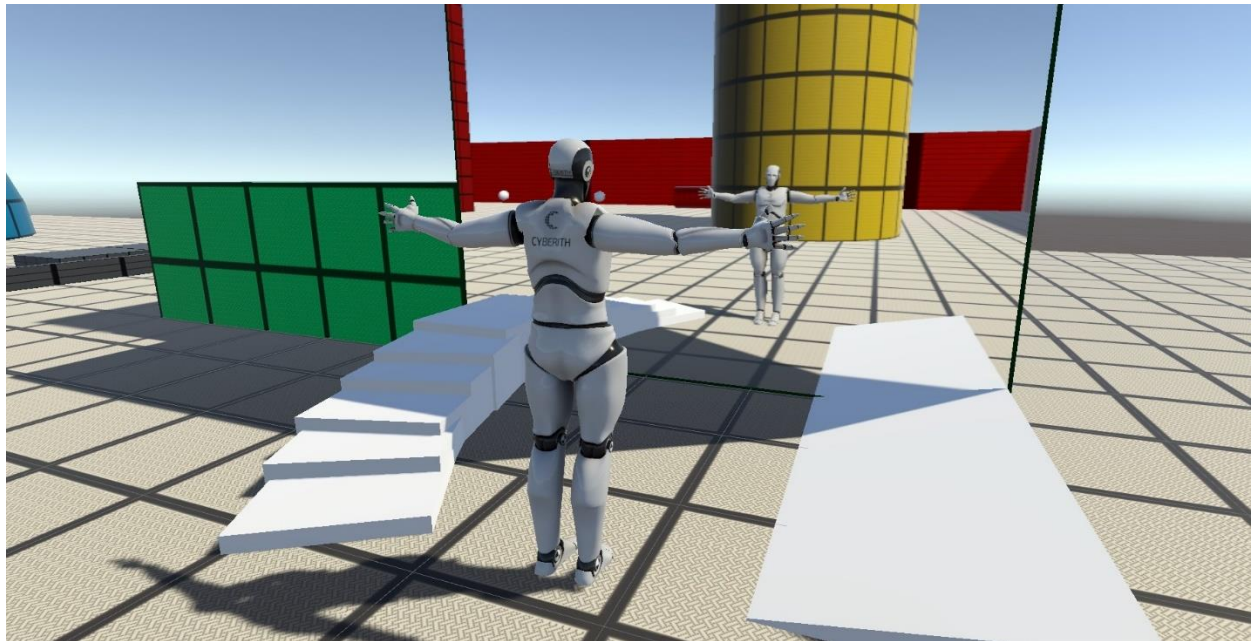
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Introduction to CyberithIK

Cyberith IK is our approach to a user friendly, powerful inverse kinematics (IK) system for realistically moving avatars in VR.

The Cyberith IK system is based on Rootmotion's FinalIK, the most powerful IK solver for Unity. As we use the IK Solver from Rootmotion, it is not an IK Solver developed by Cyberith. Instead of creating our own IK Solver, we focused on optimizing the most powerful system available for best usability for Virtualizer customers. We found that optimally implementing an advanced IK System was a challenging task for many of our customers and that the results of such IK implementation projects often did not live up to the capabilities of Final IK and the Virtualizer. We created CyberithIK to make such an implementation as easy and convenient as possible.



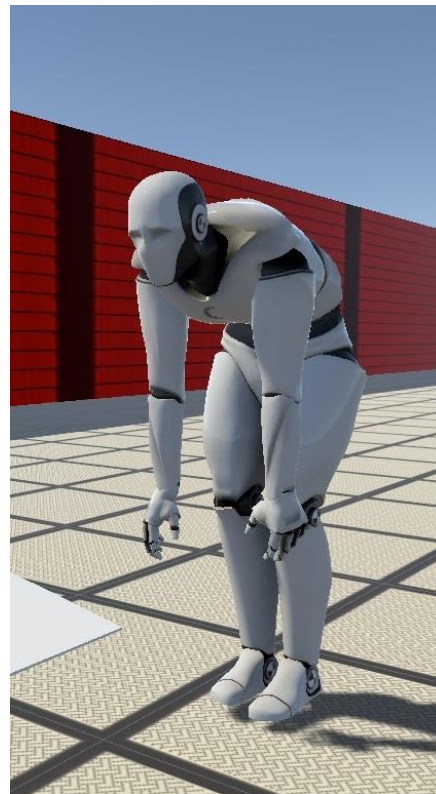
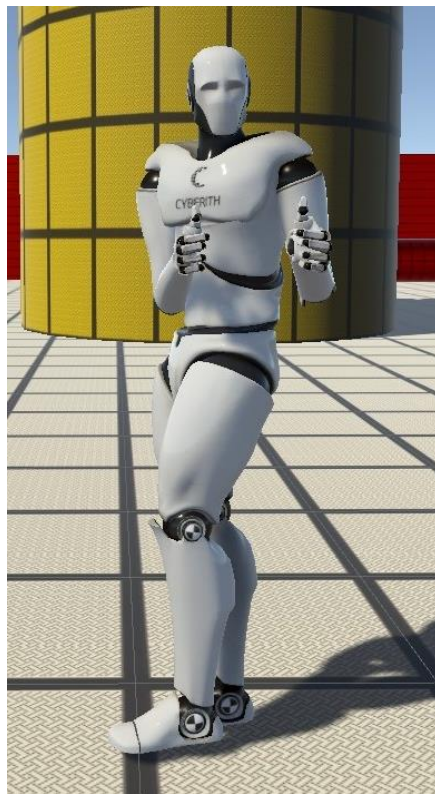
Concept of CyberithIK – What is it?

In short, we did the following main steps to create an easy to use and well working inverse kinematics (IK) System, that's based on Final IK. It adds the following main components to Final IK:

The Virtualizer

- While a typical IK System only uses 3 known points (headset position & two hand controller positions) to estimate the positions of all other joints of the body, the Virtualizer additionally provides the exact position of the hips. Thus, Cyberith IK is used as a **4-point** (headset, 2x controllers & hips) **or a 6-point IK system** (headset, 2x controllers, hips & 2x feet) – and not “just” as a 3-point IK system!

As the hips are located at a critical, central location of the human body, implementing the position of the hips into the system leads to **massively better results**. The movement of the complete torso of a person is in close relation to the movement of the hips, thus the importance of this step.



Two examples: A 3-point IK System can't determine if the user turned the body. Also, it can't differentiate between bending over and kneeling. A 4-point IK system can!

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- Additionally, the Virtualizer also provides movement data of the user:
 - In case a **4-point IK system** is used, a **procedural animation** showcases convincing movements of the legs and feet to 3rd parties. The movement of the feet is synchronized, so that it fits the movement data provided by the Virtualizer. In order to avoid confusion for the user in 1st person view, the legs are ideally hidden in that perspective. That's because the procedural animation does not match with the exact movements of the feet (e.g., which of the two feet is currently in the front). We think that the exact movement of the legs is of low importance in 3rd person view as long as the procedural animation looks realistic, because moving legs are typically not a very important part of human communication.
 - **Two SteamVR Trackers** (e.g., Vive Trackers) can be attached to the user's feet in order to use Cyberith IK as a **6-point IK system**. In that case, the exact positions of the feet will be displayed in VR correctly. Cyberith IK equally supports both the 4-point and the 6-point IK modes and detects the current mode automatically by checking if trackers with the roles "Left Foot" and "Right Foot" are currently available.

Offset Vectors

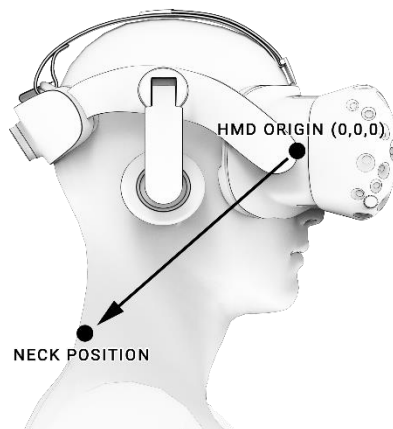
We implemented a system of offset vectors, that **connect the positions of the tracked objects with the positions of the related joints of the human body**.

Please allow us to explain this concept by an example:

In typical implementations of IK systems, the position provided by the VR headset is considered to be identical to the position of the head. In Unity, however, the position of the head (or at least what's called so) is the position of what we'd call the position of the neck of a real person. That's because the head is an extended object and the neck can be seen as its origin, the point where it is attached to the body. Thus, the nomenclature of Unity is logical, but can be confusing, as one may (wrongly) assume that the position of the head of the avatar can be connected directly to the position of the head provided by the VR headset.

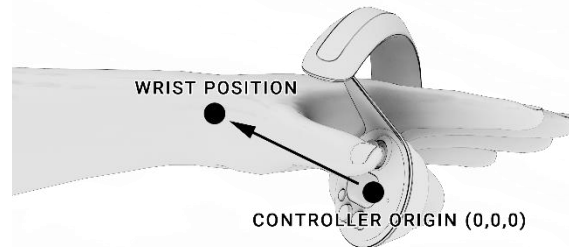
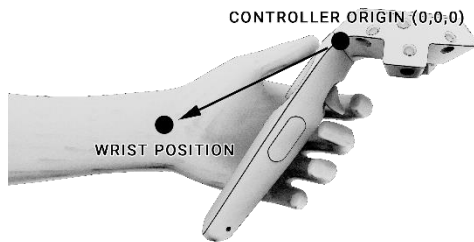
In fact, it leads to bad results if the position of the headset is connected to neck of the avatar directly, as the real person does not wear the VR headset in the neck.

Thus, we have created an offset vector (**Neck Offset**), that calculates the position of the neck based on the position of the VR headset / Head Mounted Display (HMD). As the vector is moving with the movement of the VR headset, it calculates the neck position of the user independently from the viewing direction. The image below illustrates the Neck Offset Vector.



The Neck Offset

Conceptually, the same is true for the positions of the hand controllers and the related positions of the wrists of the user. As VR hand controllers typically also have a rotational offset (in addition to the translational/linear offset), we have created two offsets: The **Hand Offset** and the **Hand Angle Offset**.



The (linear) Hand Offset for the Vive Controller ("Vive Wands") and the Index Controller ("Knuckles").

As the images above illustrate, the Hand Offsets and the Hand Angle Offsets are **different, depending on the hardware used**. Thus, you need to adapt the offsets according to your hardware setup.

Additionally to the offsets described above, CyberithIK also offers the following offsets to adjust:

- **Foot Offset and Foot Angle Offset:**
Same concept as Hand Offset and Hand Angle Offset. Describes the offset in between the Vive Tracker worn on the foot and the ankle of the user.
These offsets are only used if CyberithIK is used as 6-point IK System with trackers mounted both on the left and the right foot.
- **Foot Offset NO Trackers:**
Linear offset used to adjust the position of the feet if no trackers are used (4-point IK).
- **Hip Angle Offset:**
Used to adjust the angle of the hips. We found this to be required for most avatars.
Note: The hip does not have a linear offset. Instead, it is calibrated as explained below.

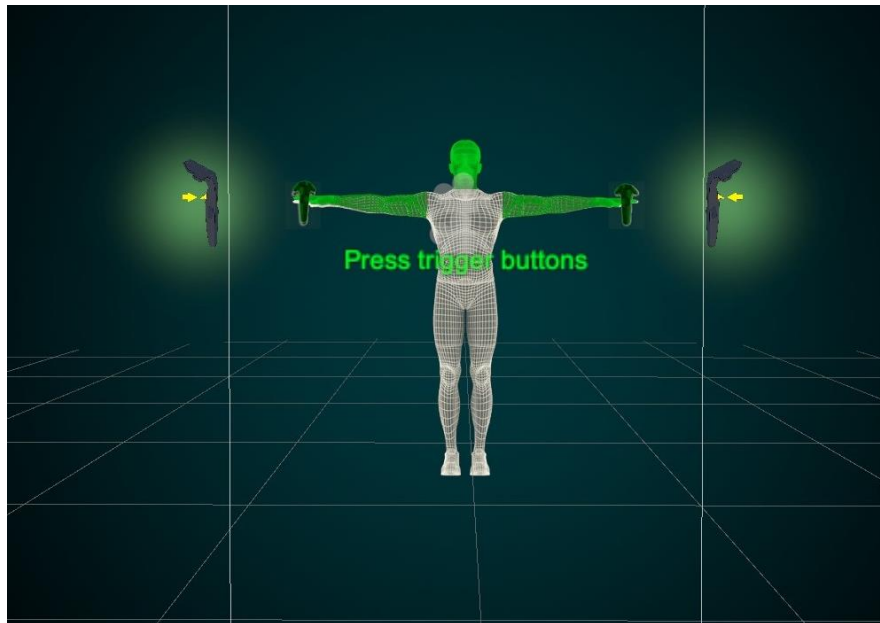
One-Step Calibration Process

We created a one-step calibration process that's extremely quick to go through and works very well for all of us.

Our calibration relies on the following **assumptions**:

- The proportions of the skeleton/bones are similar for all adult humans.
According to our experience, this assumption seems to be very good, much better than one would expect. It seems that the proportions of the bones are the same for everybody, independently of height, weight, circumference/thickness and sex.
However, it should be noted that all of us are European or "Caucasian" people in some way, so we can't be sure if the proportions are the same for people that are genetically more different. Also, please note that this assumption is not very good for children, as children have different proportions than adults.
The only part of the body, that we scale separately are the arms, as human perception of hands and arms seems to be very sensitive.
- The position of the neck defines the vertical rotational axis of the user in the Virtualizer and thus matches the center of the Virtualizer. We found this assumption to be good enough in order to receive very good results if the Neck Offset is optimized accordingly.

For calibrating, the user must **stand in the Virtualizer in T-Pose**, looking straight forwards. Once the position is correct, the user must press the trigger button on the controller to trigger the calibration.



The calibration screen

Technically, the calibration works as follows:

- It calibrates the height of the user, depending on the height of the VR headset. This step scales the complete avatar.
- It calibrates the height of the harness, as people may wear the harness higher up or lower than others.
This step of calibration uses the assumption that the height of the hip position of the avatar relates to the height of the eyes of the avatar similar to the relationship of 103 to 190. (If required, this relation can be updated directly in the script.)
- It scales all offsets, which are described above.
- Once the above mentioned steps are finished, it calibrates the length of the arms. This is done because the first assumption may not be good enough for the arms.

Requirements for achieving good results with the IK System

To achieve good results, it is of critical importance to use good sensor data and to have a precise calibration. This is both true for the Virtualizer & for SteamVR as the data of both systems is combined in the CyberithIK system.

If only one single component does not provide good data, CyberithIK might not work well.

Practically, you should check the following points:

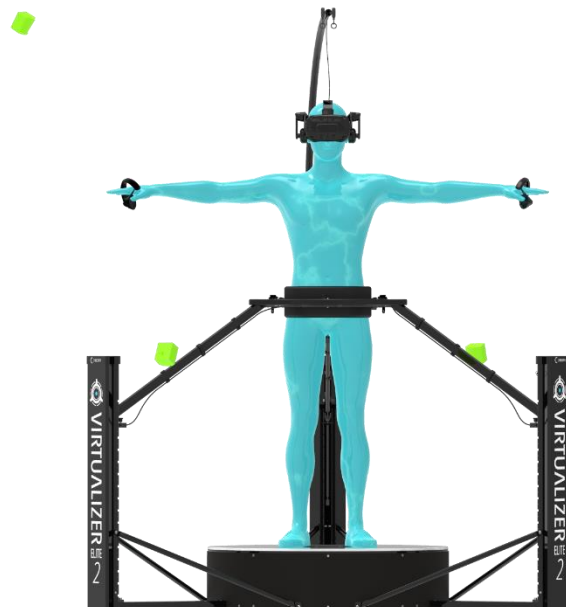
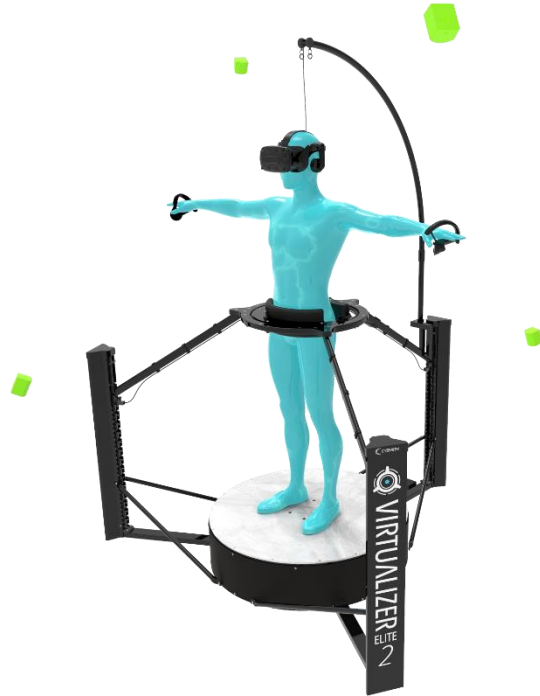
SteamVR Tracking

We are admirers of the SteamVR Tracking System with base stations / “Lighthouses” of the version 2.0. However, while SteamVR Tracking certainly is a great tracking system, there are limitations when it comes to tracking accuracy. Tracking accuracy is in the centimeters, so errors of multiple centimeters can occur. You should try to minimize these tracking errors.

Our tips:

- Make sure to have high quality tracking in your room! Correctly positioning the base stations and removing/avoiding reflective surfaces may help to improve tracking quality.
- The further away the base stations are from the tracked objects, the higher the tracking errors get. Thus, keep the base stations rather close to your Virtualizer.
- It is very beneficial to add a 3rd and ideally also a 4th SteamVR 2.0 base station to your room. Using 2 base stations is the bare minimum, which is not optimal for sensitive applications like an IK System.
- Specifically, we can recommend to add a 3rd and ideally also a 4th base station that’s close to the tracked objects in the hands and mounted on the feet of the user in the Virtualizer. In case Vive Trackers are used on the feet, it makes sense to have these additional base stations mounted at a lower height, next to the Virtualizer (rather than on the ceiling, where we’d recommend mounting the first two base stations).

The following page shows two images of how we recommend mounting the base stations. For better visibility, the base stations are shown in green.



SteamVR Room Setup

As CyberithIK combines the sensor data provided by SteamVR with the sensor data provided by the Virtualizer and also is a sensitive system, it is very important to have a precise SteamVR Room setup. It may be required to re-do the SteamVR Room Setup more regularly if CyberithIK is used.

Please follow the applicable [instructions document](#) (“Virtualizer + HTC Vive / Valve Index Setup”) to do the SteamVR Room Setup in a precise way. For most precise results, we recommend to place the VR headset on the floor directly as laid out in that instructions file. (You can simply take down the complete cable arm of the Virtualizer, so that you don’t need to unthread the headset’s cable from within the cable arm.)

IMPORTANT NOTE:

CyberithIK re-calibrates the center in its calibration phase, so **slight** offsets in the calibration of the center can be tolerated. However, the **correct calibration of the floor height** (that must match the height of the Virtualizer’s base that the user walks on) **and of the forwards direction are critical!**

Incorrect calibration of the floor height leads to wrong measurement of the user’s height, which in turn leads to a wrongly scaled avatar.

Aside of causing wrong walking directions (as always!) an incorrect calibration of the forwards direction, leads to a slightly twisted or twined avatar.

Virtualizer

CyberithIK used both the **ring angle** (“ring”) and the **ring height** (“height”) values of the Virtualizer. Make sure that both values are working properly by checking them in the Virtualizer Control Panel.

If required, you can re-calibrate the Virtualizer by following this procedure (laid out in detail in the applicable instructions file “[Virtualizer Calibration Instructions](#)”):

- Unplug the Virtualizer’s USB cable
- Plug it back in after a few seconds of waiting
- Directly after plugging in the USB cable:
 - Turn the ring multiple times
 - Move the ring up and down multiple times

Lastly, it is important that the harness is tightened properly, so that it can’t slip up or down while being in the VR experience!

The Layer System & the Avatars of CyberithIK

The “LocalPlayer” & the “RemotePlayer” Layers

Ideally, an IK System as CyberithIK is used in a multi-user setting. Thus, it requires **two layers**, the “**LocalPlayer**” and the “**RemotePlayer**” layer.

The “LocalPlayer” layer is the one visible to the user of the avatar himself (1st person), while the “RemotePlayer” layer is how a 3rd person (another VR user, a spectator, etc.) sees the avatar of the user. The mirror implemented in the CyberithIK Plugin also uses the 3rd person view (“RemotePlayer”).

This distinction in between the 1st person and the 3rd person view is required for two reasons:

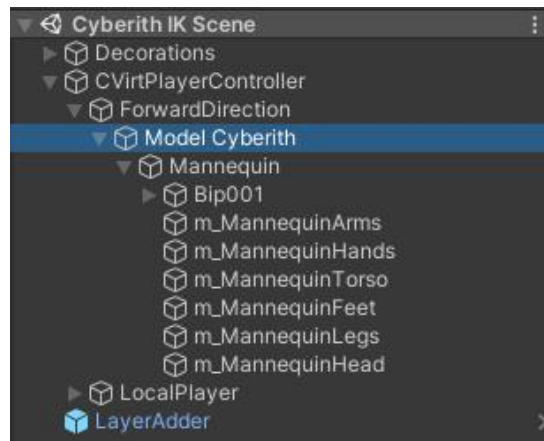
- **The Head:**
 - In 1st person view, the head should be hidden in order to avoid the possibility of the user to see the head (or the nose, etc.) from the inside!
Seeing the head from the inside looks horrible and may block the user to see the surrounding.
 - In 3rd person view (to other user’s, to the mirror, etc.), however, the head should be visible – at least in case you don’t want to create a zombie game.
- **The Legs & the Feet in case 4-point IK is used:**
 - In 4-point IK mode (without SteamVR Trackers, as explained [before](#)) the legs follow a procedural animation, which looks realistic & nice to 3rd person viewers. Thus, it should be visible in 3rd person view.
 - However, having virtual legs that follow an animation rather than representing the positions of the real legs, can be disturbing to some users. Thus, we recommend hiding the legs in 1st person view if possible.
 - This does not apply if 6-point IK is used! In that case, the feet and legs move realistically and should thus be shown both in 1st person and 3rd person view.

The Cyberith Avatar

CyberithIK comes with an optimized avatar, the **Cyberith Avatar** (“Model Cyberith”). That’s the avatar that’s visible on prior pages of this document.

The Cyberith Avatar is characterized by the following aspects:

- **The proportions are realistic**, a requirement of all avatars used for IK purposes.
- **Individual parts of the body have separate mesh renderers**, so that they can be displayed (or hidden) separately through the layer system:
 - This allows the head to be hidden in the “LocalPlayer” layer, while being shown in the “RemotePlayer” layer.
 - Optionally, it allows to do the same for the feet and legs.
- The **skeleton / bone structure follows a standard**, that’s also used by **Final IK’s** sample avatar (the famous crash test dummy) and the **Microsoft Rocketbox Avatars**. This allows CyberithIK to properly work without requiring modification with other avatars that support the same standard.



The Unity screenshot shows the separated mesh renderers for the arms, the hands, the torso, the feet, the legs and the head of the Cyberith Avatar.

The Microsoft Rocketbox Avatar Library

The **Microsoft Rocketbox avatar library** is a collection of **115 avatars** representing humans of different genders, races and occupations, **that's free to use for research and academic use!**

You can find more information here: <https://www.microsoft.com/en-us/research/blog/microsoft-rocketbox-avatar-library-now-available-for-research-and-academic-use/>

All avatars can be downloaded from Microsoft's GitHub account:
<https://github.com/microsoft/Microsoft-Rocketbox>

If you didn't take a look at these avatars already, we strongly recommend you to do so. The library **offered by Microsoft for free (!)** contains firemen, construction workers, medical doctors and surgeons, military personnel, police officers, businesspeople, football players, an airline pilot, and many more.

The CyberithIK Plugin contains a sample scene using a Rocketbox avatar (Examples/Cyberith IK Rocketbox Scene).



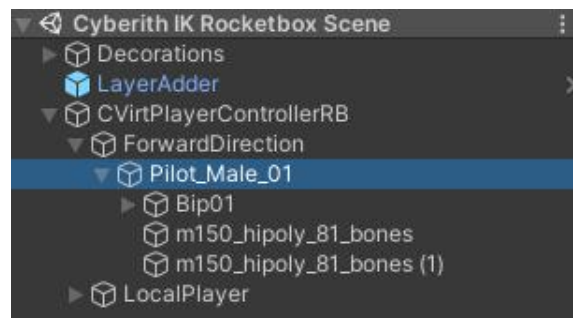
The Microsoft Rocketbox Avatar in the sample scene (airline pilot).

All Microsoft Rocketbox avatars that represent adults follow the exact same technical structure. They are characterized by the following aspects:

- The **bone structure/skeleton is identic to the one of the Cyberith Avatar**, making it easy to switch in between the Cyberith Avatar and all MS Rocketbox avatars. The proportions are matching those of the Cyberith Avatar as well.
- Additionally, the Rocketbox Avatars also **have a bone structure within the head, making them compatible with face-tracking!** That feature is specifically interesting if you have a face-tracker (e.g., HTC Vive Face Tracker) and an eye-tracking VR headset (Vive Pro Eye, Varjo, etc.)
- Unfortunately, the Rocketbox Avatars **don't have separated mesh renderers** for various parts of the body: The Rocketbox Avatars have only one Mesh Renderer, but the material of the mesh is separated in a body part and head part.

This causes a difference for the feet & legs, in case 4-point IK is used (no difference for 6-point IK):

- It is not possible to show the animation to 3rd person viewers while completely hiding the legs to 1st person viewers.
- Optionally, it is possible to insert a horizontal Clipping Plane Shader to the at the height of the hips, that hides everything below that height and thus hides the legs and feet. However, this has the severe disadvantage that it also hides other parts of the body if they get below the level of the Clipping Plane Shader. Practically, it means that the user won't be able to see his hands if he bends down and moves his hands below the level of his hips. Thus, we consider this option to be experimental.
- Technically, the models can be modified with a software tools like Autodesk Maya or Blender, but that requires additional efforts.
- Due to the points stated, adding SteamVR Trackers/Vive Trackers to the feet might add more significant value if you use Microsoft Rocketbox avatars. If you use the Cyberith Avatar, showing realistically moving legs (rather than an animation) may be a nice plus, but the added value is debatable for many applications.



The Rocketbox Avatars have only one Mesh Renderer, but the material of the mesh is separated in a body part and head part. This allows for making the head transparent by duplicating the Mesh Renderer, as explained in [the instructions for the setup of a custom avatar](#).

For 6-point IK only: The Feet Trackers

Assembling and Mounting the Feet Trackers

While elaborate explanations are not required on how to wear a VR headset and how to hold VR hand controllers correctly, mounting StreamVR Trackers onto the feet is not as simple. It requires additional components (the mounts), correct positioning on the feet and correctly setting up the tracker roles in the SteamVR settings.

Please follow the instructions provided in the according document in order to mount the Feet Trackers correctly onto your feet!



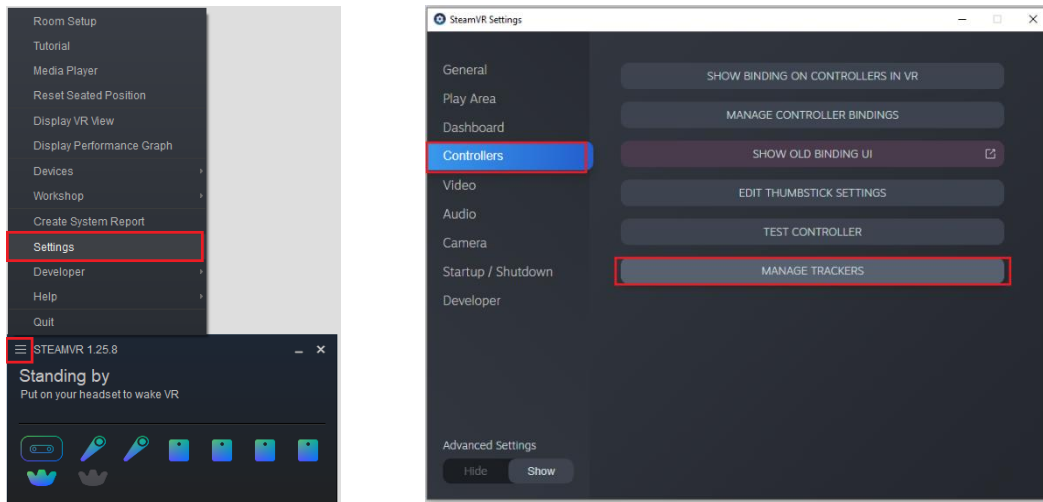
Make sure to wear the Vive Trackers and Overshoes as shown in this photo.

Note: The buckles of the straps must be on the outsides of the feet.

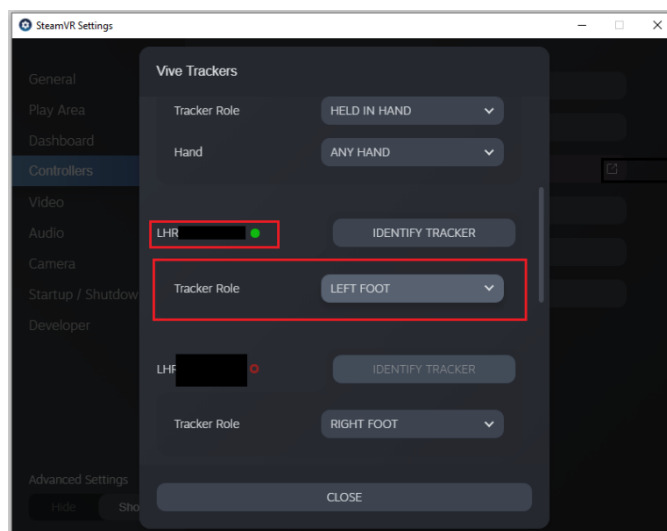
Correctly setting up the Tracker Roles in SteamVR (“Left Foot” & “Right Foot”)

The Roles of the Trackers must be set to “Left Foot” and “Right Foot” in SteamVR’s menu.

- 1.) Turn on and pair the tracker mounted on your left foot
- 2.) Go to settings and then to Controllers/Manage Trackers



- 3.) Search for the tracker currently turned on (green dot) and select “Left Foot” as the Tracker Role.



- 4.) Repeat the step for the tracker mounted on the right foot.

After going through this document, you are ready to try the Cyberith IK Demo application or to continue with the Integration Guideline for the Unity Plugin.

Important Note:

The Cyberith IK Demo application has been optimized for HTC Vive Pro (2) and Vive Controllers (“Vive Wands”). Using different SteamVR compatible hardware works too, but the results of the IK system may be less optimal! (The positions of the joints of the avatar may be slightly offset.)